Usage simulation

Where are the codes and what they do

Before running the simulation, it helps to know where everything lives. We have organized the project into a modular architecture, meaning the physics math is separated from the scripts that actually run the simulation.

```
molecular-dynamics-of-water/
├── scripts/ # 🟢 YOU ARE HERE: Scripts to run the simulations
│   └── simulate_water_cluster.py # Main runner script (edit variables here)
│
├── src/mdwater/ # the core
│   ├── topology.py # Builds the initial water box and molecule shapes
│   ├── bonds.py # Math for internal O-H stretching forces
│   ├── angles.py # Math for internal H-O-H bending forces
│   ├── lennard_jones.py # Math for short-range van der Waals forces
│   ├── coulomb.py # longrange electrostatics(Vacuum, Ewald, PME)
│   ├── force_field.py # Sums all forces/energies from the files above
│   └── velocity_verlet.py # The Integrator
│
└── src/mdwater/utils/ # HELPERS: Tools for exporting data and making plots
    ├── exporting.py # Logic for writing .xyz trajectories and logs
    └── plotting.py # Logic for generating the energy/distance graphs
```

Running Your First Simulation

Please check the number of molecules before, it might be N too large and take a lot

`scripts/simulate_water_cluster.py`. This file acts as the "control panel" for the physics engine.

To run a simulation, ensure your virtual environment is activated, (and that you are on the branch that contains code: develop for now) then type:

```bash
python scripts/simulate_water_cluster.py
```

When the simulation finishes, a Matplotlib window will pop up showing the results of the system. Close the window to complete the script.

What did this just generate?

If you have `SAVE_DATA = True` set in the script, it will generate several files in your main project folder:

1. `.xyz` Trajectory File: Contains the 3D coordinates of every atom over time. You can drag and drop this file into software like VMD or Ovito to watch a 3D animation of the water!
2. `_log.txt` File: A text file containing a summary of the simulation parameters (box size, time steps, physics mode).
3. `.png` Plot Files: High-quality images of the energy conservation graphs, ready to be pasted into our presentation!

   Note: If `SAVE_DATA = False`, the script will NOT generate these files. It will simply run the math and show you the final plots on your screen at the end of the run. This is great for quick testing!

How to Customize the Simulation

If you open `scripts/simulate_water_cluster.py` in your text editor, look at the very top under `# 1. INITIALIZE THE SIMULATION`. You can change these variables to run different types of experiments.

1. Changing System Size & The 2-Molecule Mode

```python
NUM_MOLECULES = 216
CREATE_HYDROGEN_BOND = True
```

- For bulk water: Set this to `64`, `125`, or `216`.
- For testing: Set this to `2`.

  The 2-molecule mode When running just 2 molecules, the code generates an extra plot showing Potential Energy vs. Oxygen-Oxygen (O-O) Distance.

  You can also toggle the `CREATE_HYDROGEN_BOND` flag:
- `True`: Flips one molecule so its Hydrogen points directly at the other's Oxygen, simulating an ideal Hydrogen bond approach.
- `False`: The molecules start perfectly parallel to each other (Oxygens facing Oxygens and Hydrogens facing Hydrogens). This allows us to study non-ideal, repulsive interaction dynamics.

2. Changing the Physics Math (Electrostatics & Box Size)

```python
ELECTROSTATICS_MODE = "PME"
```

For calculating the long-range electric forces you can switch between three modes: `"VACUUM"`, `"EWALD"`, or `"PME"`.

- `VACUUM` all $\mathbb{R}^3$, no PBC, no cutoff, electrostatic is calculated with Coulomb as normally
- `EWALD` PBC, cutoff and standard Ewald Summation
- `PME` Improves EWALD by using fast fourier transform.

  The Box Size Logic:

  There is a block that automatically calculates the `current_box` and `current_cutoff` based on the number of molecules and the volume of each one (30 Å^3).

  The minimum values are `current_cutoff=9.2` and `current_box=20`, because Lennard-Jones potential vanished mostly at 9 Å, (and let's give something extra). And the cutoff ratio has to be less than or equal half of the box size, (for minimum image convention MIC).

3. Adjusting the Time and Length

```python
AKMA_TIME_TO_FS = 48.88
target_step_fs = 0.25
num_steps = 12000

dt = target_step_fs / AKMA_TIME_TO_FS
```

- AKMA Our physics model runs on AKMA units (Angstroms, Kcal/mol, Mass in AMU). Because of this dimensional analysis, 1 unit of time corresponds to `48.88` femtoseconds.
- If you want a quick test run, drop `num_steps` down to `1000`.

Examples of some simulations

2 molecules

- 👀 For checking the exact values of variables like Lennard-Jones constants A,B or C for Coulomb, and equilibrium values, please see `src/mdwater/force_field.py`

  (We are trying to see the Hydrogen bounds acting)

  INPUT 1.

```python
# ==========================================
# 1. INITIALIZE THE SIMULATION
# ==========================================
NUM_MOLECULES = 2  # Set to 2 to test the hydrogen bond configuration and potential bowl; increase for larger clusters
SAVE_DATA = True # TO save the trajectory, plots energy and logs

# --- NEW: The Electrostatics Switch ---
# Options: "VACUUM", "EWALD", "PME"
ELECTROSTATICS_MODE = "VACUUM"  # Change this to test different electrostatics modes (VACUUM, EWALD, PME)

spacing = 3.1 #standard separation
```

```python
# --- 2-MOLECULE mode ---
CREATE_HYDROGEN_BOND = True
p0 = np.zeros_like(q0)
```

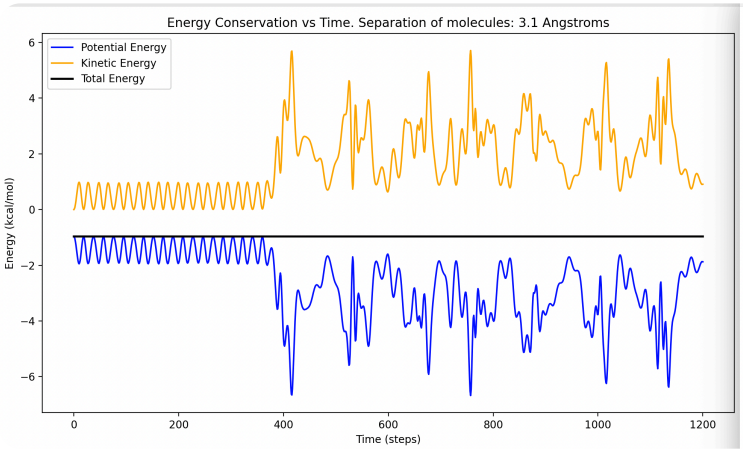
```python
# ==========================================
# 2. RUN INTEGRATOR. OPTIONAL TRACKING
# ==========================================
AKMA_TIME_TO_FS = 48.88
target_step_fs = 0.25
num_steps = 12000
dt = target_step_fs / AKMA_TIME_TO_FS
TRACK_INTERVAL = 10 # Tracks energy every 10 steps to avoid tracking every single step
```

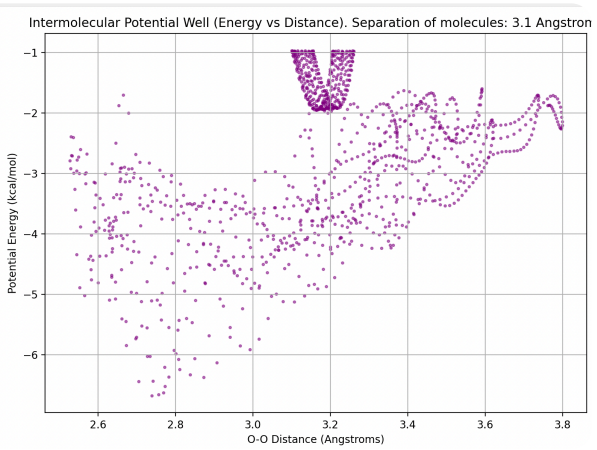
Output 1

```bash
==============================================
    MDWATER SIMULATION CONDITIONS
==============================================
 MODE:           VACUUM
 BOX SIZE:       Infinite
 CUTOFF:         None
```

```
 ELECTROSTATICS: Direct Coulomb
==============================================

Starting simulation for 12000 steps...
Simulation complete!
Saved energy plot to 2_water_vacuum_time_plot.pdf
Saved distance plot to 2_water_vacuum_bowl_plot.pdf
Exporting trajectory to 2_water_vacuum.xyz...
Export complete!
Writing initial conditions to 2_water_vacuum_log.txt...
Log saved successfully!
Saved energy plot to 2_water_vacuum_time_plot.pdf
Saved distance plot to 2_water_vacuum_bowl_plot.pdf
```

And the plots are:



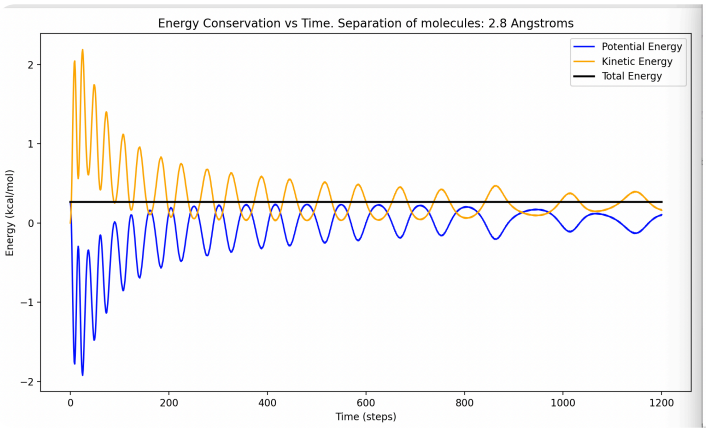


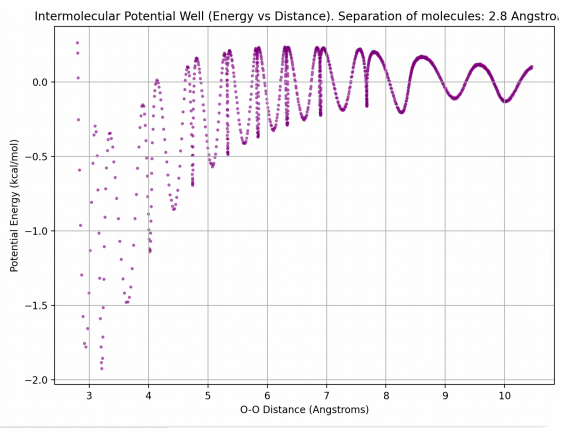
INPUT 2

- If we keep all variables except the separation between molecules:

  `spacing = 2.8`

  ### OUTPUT 2 (plots)



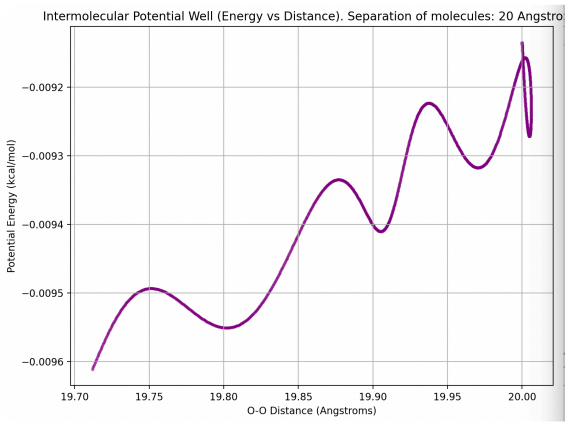


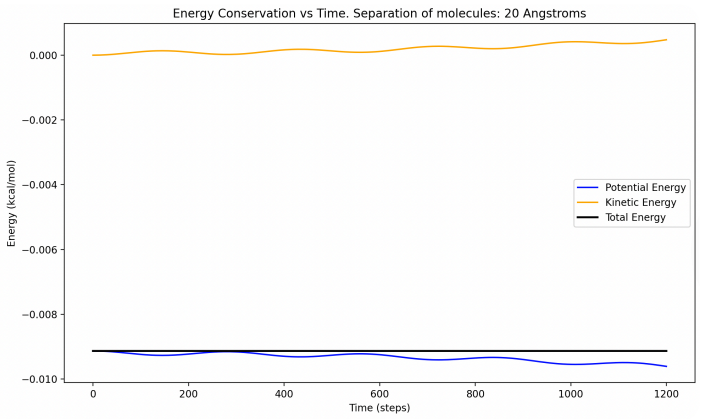
INPUT 3

- If we separate them too much:

  `spacing = 20`

OUTPUT 3 (plots)





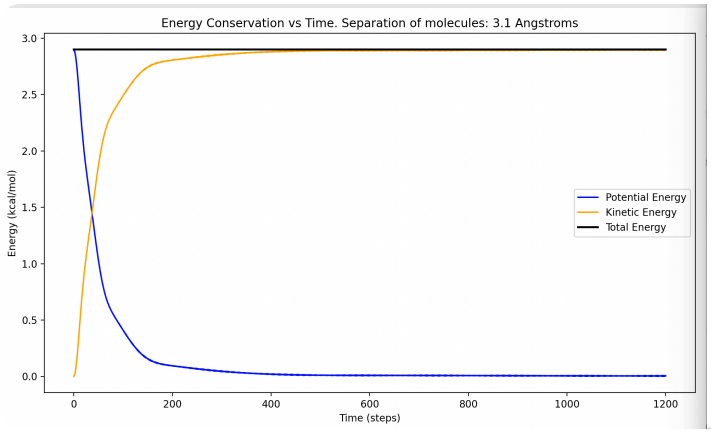
INPUT 4

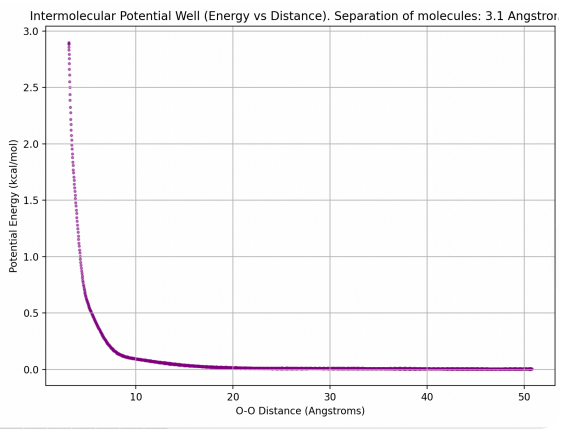
- If we keep the first value for separation but we turn off the hydrogen bounds, molecules are facing parallel, so they repel (we expect)

  `CREATE_HYDROGEN_BOND = False`

  `spacing = 3.1`

OUTPUT 4 (plots)





yes, they go away from each other.

A cluster of molecules

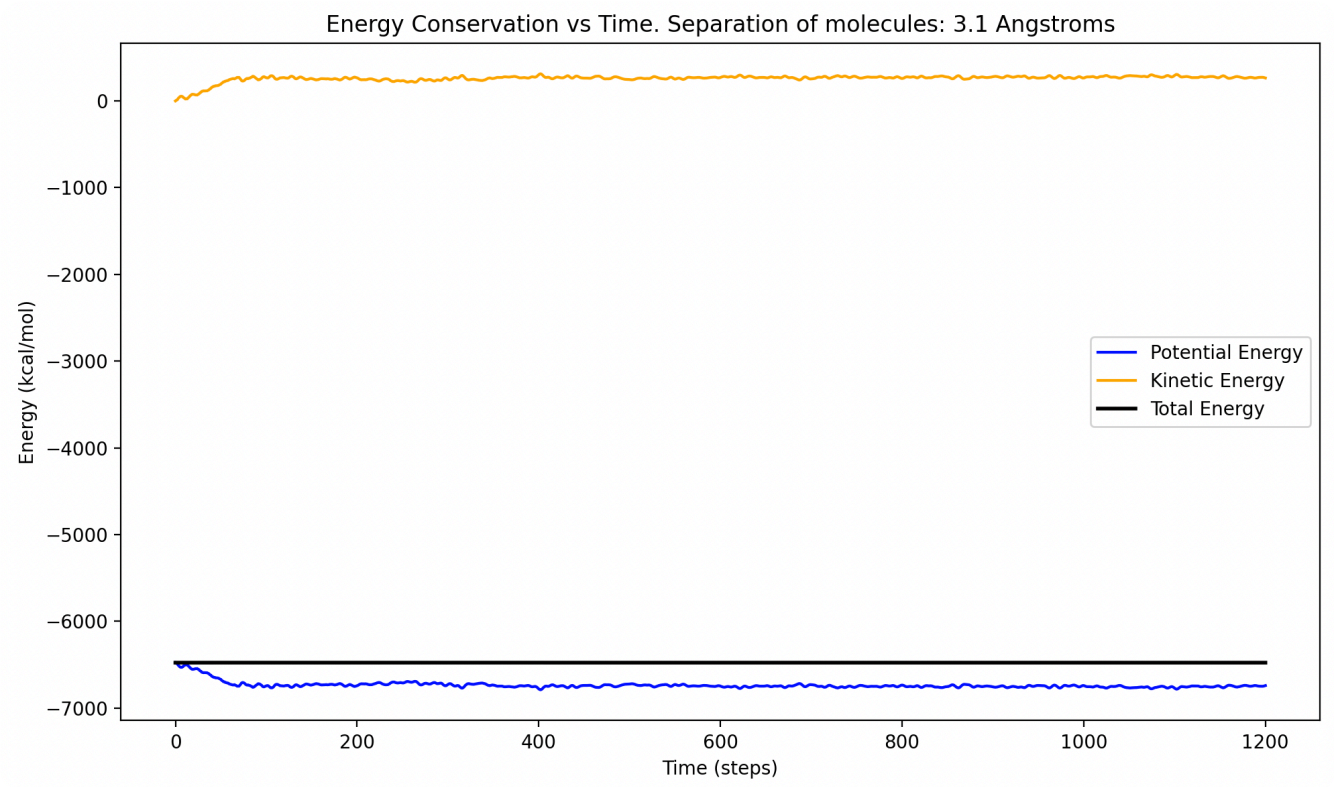
Setting to more molecules:

INPUT 5

```python
NUM_MOLECULES = 100  # Set to 2 to test the hydrogen bond configuration and potential bowl; increase for larger clusters
SAVE_DATA = True # TO save the trajectory, plots energy and logs

# --- NEW: The Electrostatics Switch ---
# Options: "VACUUM", "EWALD", "PME"
ELECTROSTATICS_MODE = "EWALD"  # Change this to test different electrostatics modes (VACUUM, EWALD, PME)
```

OUTPUT 5

```bash
Building universe with 100 water molecules...

==============================================
    MDWATER SIMULATION CONDITIONS
==============================================
 MODE:           EWALD
 BOX SIZE:       20.0 Å
 CUTOFF:         9.2 Å
 ELECTROSTATICS: EWALD Summation
==============================================

Starting simulation for 12000 steps...
Simulation complete!
Saved energy plot to 100_water_pbc_EWALD_time_plot.pdf
Exporting trajectory to 100_water_pbc_EWALD.xyz...
Export complete!
Writing initial conditions to 100_water_pbc_EWALD_log.txt...
Log saved successfully!
Saved energy plot to 100_water_pbc_EWALD_time_plot.pdf #🤣 it prints twice
```



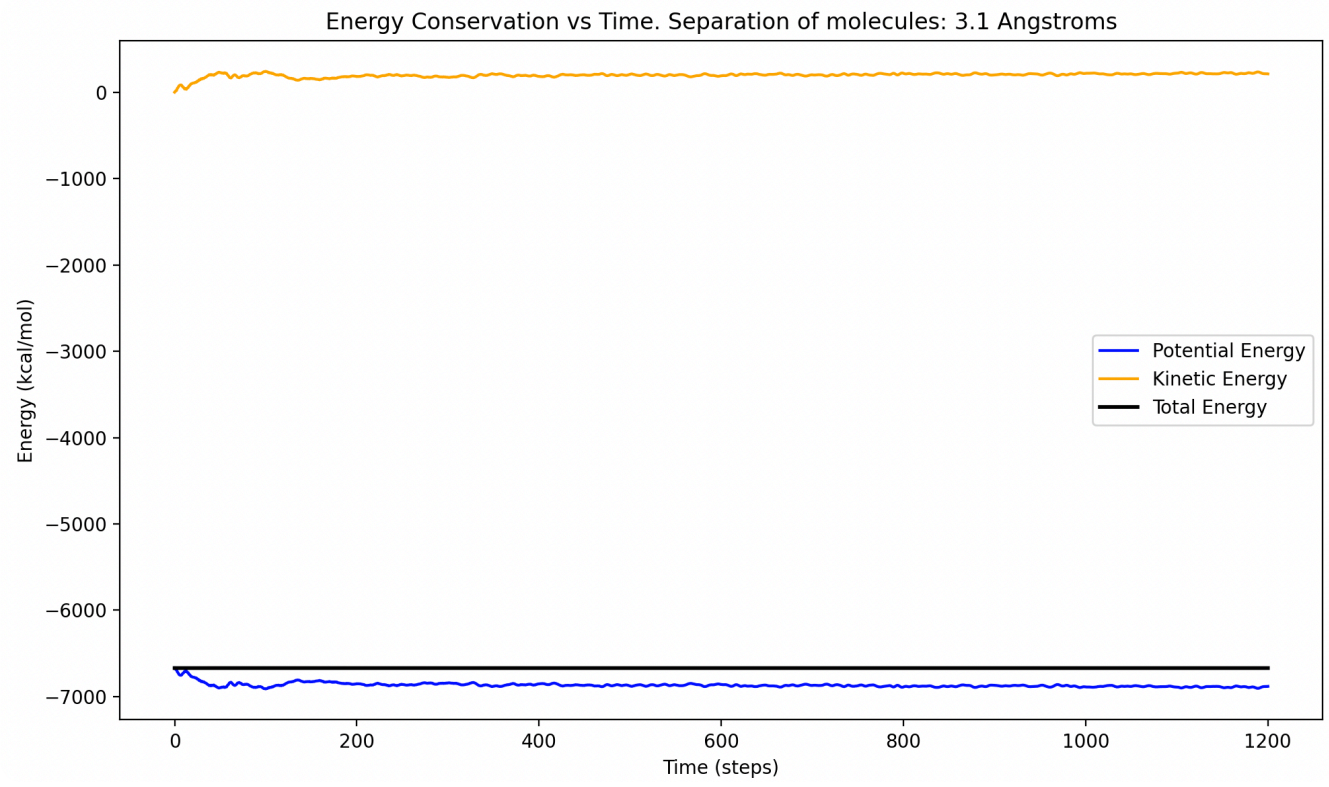
INPUT 6

- Changing the mode for PME, we should get the same.

  `ELECTROSTATICS_MODE = "PME"`

  OUTPUT 6

```bash
Building universe with 100 water molecules...

==============================================
    MDWATER SIMULATION CONDITIONS
==============================================
 MODE:           PME
 BOX SIZE:       20.0 Å
 CUTOFF:         9.2 Å
 ELECTROSTATICS: PME Summation
==============================================

Starting simulation for 12000 steps...
Simulation complete!
Saved energy plot to 100_water_pbc_PME_time_plot.pdf
Exporting trajectory to 100_water_pbc_PME.xyz...
Export complete!
Writing initial conditions to 100_water_pbc_PME_log.txt...
Log saved successfully!
Saved energy plot to 100_water_pbc_PME_time_plot.pdf
```



DataView Error

```
Dataview (inline field '==============================================
    MDWATER SIMULATION CONDITIONS
================================================
 MODE:           VACUUM
 BOX SIZE:       Infinite
 CUTOFF:         None
 ELECTROSTATICS: Direct Coulomb
================================================

Starting simulation for 12000 steps...
Simulation complete!
Saved energy plot to 2_water_vacuum_time_plot.pdf
Saved distance plot to 2_water_vacuum_bowl_plot.pdf
Exporting trajectory to 2_water_vacuum.xyz...
Export complete!
Writing initial conditions to 2_water_vacuum_log.txt...
Log saved successfully!
Saved energy plot to 2_water_vacuum_time_plot.pdf
Saved distance plot to 2_water_vacuum_bowl_plot.pdf'): Error:
-- PARSING FAILED ---------------------------------------------------

> 1 | ==============================================
    | ^
  2 |     MDWATER SIMULATION CONDITIONS
  3 | ================================================

Expected one of the following:

'(', 'null', boolean, date, duration, file link, list ('[1, 2, 3]'),
negated field, number, object ('{ a: 1, b: 2 }'), string, variable
```